Design

Architecture

main.py provides the command line. src/calculations.py contains side-effect-free Decimal formulas. src/engine.py validates CSV/JSON, writes source records, closes disputes, builds weekly statements, and exports reports. database/database.py opens SQLite and applies database/schema.sql on each connection. The existing database file is database/settlements.sqlite3.

Processing flow

- Register restaurant profiles.
- Import orders (including optional item rows and dispute status). A SHA-256 history entry makes identical file imports idempotent.
- Inspect `order-breakup` before settlement.
- Open disputes remain unassigned. CSV dispute events can open or close a dispute; an explicit supported refund amount can update the order refund ledger.
- Weekly settlement selects unassigned eligible orders, calculates one statement, records a complete per-order audit snapshot, then locks those orders to its unique statement.
- A post-settlement dispute change uses `correct-settlement`. The original statement and audit are retained; a new revision and correction audit are appended. Reports display the latest statement revision and both audit histories.
- Reports export calculated order breakups, current settlement revisions, and audit history as CSV or JSON.
- Imports validate required fields and numeric flags before commit. Duplicate order IDs within a batch fail the transaction. Failed dispute batches also roll back as a unit. A repeated correction with no changed monetary result does not append a duplicate revision.

Money and reporting

All money values are Decimal text in SQLite and quantized to two decimal places with ROUND_HALF_UP. TDS GMV is iterated in chronological order without converting to binary floating point. Commission uses gross less restaurant-funded discount; packaging is separate. A sub-Rs 100 order pays the fixed platform fee instead of commission. Section 9(5) GST is displayed as a deduction because the platform withholds and remits it for the restaurant. Delivery fees are totalled for visibility but excluded from payout pending a specified recipient/funding rule; see the README for assumptions.

Inputs

Restaurant columns:
restaurant_id,name,plan,onboarding_date,rating,packaging_charge,weekly_ad_budget (optional status).
Order columns: order_id,restaurant_id,order_time,items_gross plus optional discounts, packaging, distance, mode, outcome, refund fields, dispute fields, cancel seconds, accepted flag, delivery fee, platform fee, rating. Itemized CSV repeats order fields on each line and uses item_name,item_quantity,item_unit_price; aggregate CSV remains compatible. JSON order objects may

add items:[{name,quantity,unit_price}]. Dispute CSV events use dispute_id,order_id,event,event_at,resolution,refund_amount,refund_reason.

Known scope choices

Rates and GST bases follow the supplied brief, with unresolved cancellation tax treatment documented separately. Platform fee below Rs 100 uses the specified flat minimum; other platform fees are accepted as already-calculated amounts. Delivery fees are recorded but not allocated without a defined recipient. Rating averages from the prior week set the following week's commission rate. For SPILLED refunds, the platform receives half rounded half-up to paise and the restaurant receives the remainder. See decision_log.md for exact interpretations and correction limits, and br_traceability.md for rule-to-test links.